

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT				1. CONTRACT ID CODE		PAGE OF PAGES 1 12	
2. AMENDMENT/MODIFICATION NO. 0004		3. EFFECTIVE DATE 24-Sep-2025		4. REQUISITION/PURCHASE REQ. NO. 0012296018-0001		5. PROJECT NO.(If applicable)	
6. ISSUED BY ARMY CONTRACTING COMMAND (ACC) REDSTONE/CCAM-MZD-B BLDG 401 VICTORY BLVD FORT EUSTIS VA 23604-5577		CODE W911W6		7. ADMINISTERED BY (If other than item 6) See Item 6			
8. NAME AND ADDRESS OF CONTRACTOR (No., Street, County, State and Zip Code)				X		9A. AMENDMENT OF SOLICITATION NO. W911W625R0006	
				X		9B. DATED (SEE ITEM 11) 03-Jul-2025	
						10A. MOD. OF CONTRACT/ORDER NO.	
						10B. DATED (SEE ITEM 13)	
CODE		FACILITY CODE					
11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS							
☒ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of offer ☐ is extended, ☒ is not extended. Offer must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended by one of the following methods: (a) By completing Items 8 and 15, and returning _____ copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.							
12. ACCOUNTING AND APPROPRIATION DATA (If required)							
13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACT/ORDERS. IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.							
A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A.							
B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(B).							
C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:							
D. OTHER (Specify type of modification and authority)							
E. IMPORTANT: Contractor ☐ is not, ☐ is required to sign this document and return _____ copies to the issuing office.							
14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.) See page 2.							
Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.							
15A. NAME AND TITLE OF SIGNER (Type or print)				16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print)			
				TEL: _____ EMAIL: _____			
15B. CONTRACTOR/OFFEROR		15C. DATE SIGNED		16B. UNITED STATES OF AMERICA		16C. DATE SIGNED	
_____ (Signature of person authorized to sign)				BY _____ (Signature of Contracting Officer)		24-Sep-2025	

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES

SECTION SF 30 - BLOCK 14 CONTINUATION PAGE

The following have been added by full text:

AMD 0004

1. The above numbered Broad Agency Announcement (BAA) is hereby amended as follows:
 - a. Attachment 2, Two-Step Process – Concept Papers, paragraph B, Research Project Description is revised to delete reference to “12-point font”.
 - b. Attachment 5, Topics/Areas of Interest, Sequence #25-ATTACH5-A is modified to:
 - (1) Revise the Description paragraph to include adding Engines: Objectives.
 - (2) Revise the Schedule paragraph to include specific concept paper submission guidance for Drives and Engines.
 - (3) Segregate the topic descriptions by Drives and Engines.
 - (4) Add the following Topic: Lightweight and High Strength Compressor Materials (under Engines Topic Descriptions).
 - (5) Add the following Topic: Optimized Compressor Aerodynamics (under Engines Topic Descriptions).
 - (6) Add the following Topic: Advanced Combustion Systems (under Engines Topic Descriptions).
 - (7) Add the following Topic: High Temperature Tolerant Turbine Materials and Durability Improvements (under Engines Topic Descriptions).
 - (8) Add the following Topic: Advanced Manufacturing to Improve Turbine Performance (under Engines Topic Descriptions).

SECTION L - INSTRUCTIONS, CONDITIONS AND NOTICES TO BIDDERS

The following have been modified:

ATTACH 2 TWO STEP**ATTACHMENT 2****Two-Step Process - Concept Papers****Step One:**

The Government will initiate the Two-Step Process by amending the BAA and adding a research and development project to Attachment 5. Interested sources will submit Concept Papers electronically using the Solicitation Module within the Procurement Integrated Enterprise Environment (PIEE) (<https://piee.eb.mil/>).

Offerors will use the following format for the Concept Papers, subject to any changes in Attachment 5 for the specific research and development project:

Section A: Cover Page. This section is one page and includes the BAA number and Title of Research and Development Project name from Attachment 5, Name of Company, Business Size, Company's Commercial and

Government Entity (CAGE) number, Unique Entity Identifier (UEI) number, Contracting and Technical POC. Offerors will not include any other information on the Cover Page.

Section B: Research Project Description. This section is limited to five single-sided 8 ½ x 11-inch typed pages (including charts, graphs, photographs, etc.). Offerors shall include a discussion of the nature and scope of the research project and the proposed technical approach/solution including identification of major technical risks with defined feasible risk mitigation efforts. It may also include any proposed deliverables. Pages in excess of the five-page limit will not be read or considered.

Section C: Rough Order of Magnitude (ROM) Cost. This section is limited to one page and provides a ROM cost estimate. A ROM cost is a quick, broadly scoped cost estimate that is typically completed when few details exist on a specific effort.

Based upon the following **Review Criteria**, the Government will categorize Concept Papers as either **“Of interest”** or **“Not of interest”**:

- (1) Technical merit of the proposed concept with an emphasis on feasible, achievable, and innovative solutions to achieve the project objectives;
- (2) the degree to which a defined transition path or strategy to a capability exists; and
- (3) the degree to which the proposed schedule is achievable and affordable based upon the ROM cost estimate.

The Review Criteria are of equal importance.

During the Government review of Concept Papers, the Government evaluators may elect to ask questions of individual offerors to enhance the Government’s understanding of and to facilitate the Government’s evaluation of the Concept Paper. The Government evaluator questions may be communicated either via the Government version of Microsoft Teams or by electronic mail.

Step One concludes with the Government Contracting Officer/Agreements Officer notification to Offerors that either:

- The Concept Paper is “Of Interest” and will continue into Step Two;
- The Concept Paper is “Of Interest, Currently Not Affordable” and the Government will inform the Offeror that the Concept Paper will be retained in the event funding becomes available and allowing the Government to request a proposal thereby initiating Step Two.
- The Concept Paper is “Not of Interest” with rationale for the decision. For “Not of interest” Concept Papers, this will end Government consideration of the Concept Paper.

Step Two: Applicable only to Concept Papers Determined “Of-Interest”

The Government Contracting/Agreements Officer will initiate Step Two, by sending a request for proposal to those Offerors whose Concept Papers are “Of interest”. During Step Two, all communications between the Government and the Offeror will be through the Government Contracting/Agreements Officer. The Government reserves the option to request proposals for all or a portion of the proposed research project. A request for proposal does not imply the Government has made a final funding decision.

Proposals shall be submitted in accordance with Attachment 3 and are due by the date specified in the request.

The Government will evaluate the proposal using the criteria from Step One substituting the cost proposal for the ROM cost estimate to determine if the proposed cost is reasonable and realistic. After an initial evaluation of the proposal, the Government may elect to conduct discussions with the Offeror to confirm proposal details; for example, discussions concerning the Offeror’s detailed statement of work for the project, and/or address proposal ambiguities.

At the conclusion of discussions if applicable, the Government will **categorize the proposal** as one of the following:

“Of Interest” and funding is available; or

“Of Interest”, but funding is not available. Government will ask offeror to extend the proposal validity period in event funding becomes available; or

“Not of Interest” and the Government does not intend to fund the proposal.

The Government will also consider program balance/diverse technologies in making decisions on proposals “Of Interest”.

ATTACH 5 TOPICS AREAS OF INT

Attachment 5 to BAA W911W6-25-R-0006

Topics/Areas of Interest

Sequence #25-ATTACH5-A

Supplemental Power, Efficient Engines, and Drives (SPEED)

Description: The Technology Development Directorate (TDD) is interested in the development and demonstration of rotorcraft engine and drive system technologies as part of the “Supplemental Power, Efficient Engines, and Drives (SPEED)” program. This effort will follow the two-step process described in Attachment 2 of the above numbered BAA. The Government will first review concept papers and may then follow up with full proposals. Offerors are requested that all concept papers clearly identify transition paths to existing or future planned Army rotorcraft platforms, including any applicable qualification challenges or considerations

Drives: Objectives related to rotorcraft drive systems include component life improvements, power density improvements, life cycle cost reductions, and loss of lubrication performance improvements. Specific topics of interest for drives systems include Topological Optimization for Additively Manufactured Corrosion Resistant Gearbox Housings, Lightweight Emergency Loss of Lubrication Systems, Artificial Intelligence and Machine Learning for Drive System Prognostics, Non-Press Fit or Splined Inner/Outer Race Fully Ceramic Bearings in Gearboxes, and Topological Optimization of Dynamic Drive System Components.

Engines: Objectives related to rotorcraft engines include specific fuel consumption improvements, power density improvements, sustainability improvements, and surge margin improvements. Specific topics of interest for engines systems include Lightweight and High Strength Compressor Materials, Optimized Compressor Aerodynamics, Advanced Combustion Systems, High Temperature Tolerant Turbine Materials and Durability Improvements, and Advanced Manufacturing to Improve Turbine Performance. Engine size classes of interest for this BAA include gas turbine engines in the 3,000-8,000 horsepower range for manned rotorcraft platforms, as well as engine systems applicable to group III and IV unmanned aerial systems (UAS).

Topic Audience: Small Businesses, Large Businesses

Department of the Army Research, Development, Test, & Evaluation (RDT&E) Budget Activity (BA): BA 2, Applied Research. program that is seeking technologies starting between a Department of Defense (DoD) Technology Readiness Level (TRL) 3 to 4 and ending at a TRL 5. Sources not familiar with DoD TRLs, please review the DoD Technology Readiness Assessment Guidebook published by the Under Secretary of Defense for Research and Engineering.

Program Element: PE 0602183A / *Air Platform Applied Research*

Project: DK1 / *Air Vehicle Integrated and Alternative Tech (AVIATe)*

Technology Areas: Engines, Drive Systems

Budget:

FY26	FY27	FY28	FY29	FY30	Total
\$1,500K	\$1,900K	\$2,700K	\$2,600K	\$2,700K	\$11,400K

Schedule: Multiple awards are anticipated each year, based upon availability of funds. Each award is anticipated to have a period of performance between 12-24 months.

Drives: Offerors are requested to provide concept papers for one or more of the listed drives topics of interest before 1 October of each year.

Engines: Offerors are requested to provide concept papers for one or more of the listed engines topics of interest before 1 December 2025 for anticipated FY26 awards, and then before 1 October of subsequent years.

Alterations from Standard Process: N/A

Opportunities for Fundamental Research: No. This work will require use of controlled unclassified information (CUI) and generation of CUI.

Technical POC: Matthew Clark (757-878-2400, matthew.b.clark34.civ@army.mil)

Drives Topic Descriptions

1--Topic Name: Topological Optimization for Additively Manufactured (AM) Corrosion Resistant Gearbox Housings

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Rotorcraft gearbox housings are typically fabricated using a Magnesium alloy and produced by either a casting or “hogout” process. Coatings are often used to prevent corrosion in Magnesium housings, but corrosion may still occur when the coatings are damaged by foreign objects. Aluminum is a potential alternate material, but galvanic corrosion can occur due to the use of steel bolts. The casting and hogout manufacturing processes often require increased thickness and create material waste during fabrication. The casting process also often requires several iterations to fabricate an acceptable new gearbox housing design and often causes material flaws which leads to scrapped housings.

Solution Guidelines: TDD seeks technologies related to topological optimization of additively manufactured corrosion resistant gearbox housings. Challenges that must be overcome in this area include reducing housing weight while maintaining performance requirements, reducing post processing requirements, detecting and correcting various fault types during the AM process, and preventing housing corrosion with a lightweight solution that is not easily damaged. Efforts are expected to perform component testing to demonstrate the challenges have been addressed and provide an analysis of the benefits towards an Army rotorcraft.

Anticipated benefits: Anticipated benefits for this topic include improved power density (due to the reduction in weight from topological optimization), increased component life (due to the reduction in corrosion and material flaws), and reduced life cycle cost (due to the reduction in corrosion, scrap rate, design iteration time, and required material).

2--Topic Name: Lightweight Emergency Loss of Lubrication System

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Army rotorcraft gearboxes are required to operate for 30 minutes after a loss of lubrication (LOL) event to provide sufficient time to safely land. The Army desires increased LOL emergency operation time to allow more time for pilot decision making on how and where to land, and to escape hostile territory before landing. Gear coatings can potentially provide a lightweight solution to improve LOL capability but have not yet demonstrated durability over time and may not be a sufficient solution for all configurations.

Solution Guidelines: TDD seeks technologies which improve the LOL capability of rotorcraft gearboxes, particularly using small, pressurized containers of emergency lubrication material that is expelled upon temperature rise in the gearbox. Challenges to overcome include the ability to ensure lubricant release at set elevated temperature conditions, device reliability independent of manual inspection, and optimizing the amount and location of devices for reduced weight and cost while maintaining effectiveness. Efforts are expected to perform component testing to demonstrate the challenges have been addressed as well as small scale testing in a demonstrator gearbox to assess performance.

Anticipated Benefits: Anticipated benefits for this topic include improved power density (due to reduced LOL system weight) and increased LOL time (due to improvements to the emergency LOL system).

3--Topic Name: Artificial Intelligence and Machine Learning for Drive System Prognostics**Preferred Research Focus:** Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Operation and Sustainment costs account for roughly 70% of the total life cycle cost of rotorcraft, with maintenance of drive systems being one of the top cost drivers. High maintenance costs are often driven by early removal of components well before failure. Condition indicators from current generation health management systems enable some capability to predict failure but often require decades of usage data for refinement into a useful tool for maintenance forecasting. Future Vertical Lift (FVL) rotorcraft will not have access to past failure data to initially develop condition indicators, leading to overly conservative inspection intervals for drive components early in the aircraft life cycle. Artificial Intelligence (AI) models are needed to predict drive system failures in FVL rotorcraft and eliminate overly conservative inspection intervals.

Solution Guidelines: TDD seeks the development of AI models that can integrate physics-based models with actual usage data to accurately predict failures for new aircraft. Challenges to overcome include determining the appropriate AI algorithm, collecting sufficient amounts and types of data for model training, determining appropriate sensor and usage data required, and validating accuracy of failure forecasting. Efforts are expected to combine physics-based and AI models and demonstrate their failure forecasting accuracy against a dataset of known failures.

Anticipated Benefits: Anticipated benefits for this topic include reduced life cycle cost (due to accelerated maturity of next-generation condition indicators) and increased aircraft availability (due to elimination of conservative maintenance intervals).

4--Topic Name: Non-Press Fit or Splined Inner/Outer Race Fully Ceramic Bearings in Gearboxes**Preferred Research Focus:** Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Fully ceramic bearings provide up to a 50% weight reduction compared to steel bearings, but the mismatch in coefficient of thermal expansion (CTE) between ceramic bearings and steel gear shafts causes cracking in the bearing inner ring. Tolerance rings have shown to allow some thermal mismatch expansion but are not designed for shaft side loading common in aviation transmissions. This side loading causes severe

compression of the tolerance rings, resulting in bearing and shaft misalignment and ultimately reduced reliability and component life.

Solution Guidelines: TDD seeks a novel bearing design incorporating a spliced inner and/or outer race to accommodate CTE mismatch and side loading. Challenges to overcome include maintaining required bearing tolerances, eliminating the formation of stress risers at the spline root, and identifying a method to detect non-metallic debris in the oil. Efforts are expected to perform component thermal testing to demonstrate performance.

Anticipated Benefits: Anticipated benefits for this topic include increased power density (due to reduced bearing weight) and increased LOL time and component life (due to improved durability of ceramics).

Other Information: For Offerors interested in this Topic, a Report entitled Future Advanced Rotorcraft Drive System II is available upon request and upon providing evidence of a Joint Certification Number and/or a copy of the requestor's approved DD For 2345 Military Critical Technical Data Agreement. Contact the Joint Certification Program Office at <https://www.dla.mil/Logistics-Operations/Services/JCP/> for more information.

5--Topic Name: Topological Optimization of Dynamic Drive System Components

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Dynamic drive system component design can be limited by traditional forging and machining processes. For example, traditional processes may require bolting or welding components together. These traditional processes may impact manufacturing time and cost, as well as system weight.

Solution Guidelines: TDD seeks topological optimization with new manufacturing processes such as additive manufacturing, which enables more complex geometries as well as reductions in cost and weight. Challenges to overcome include developing an optimized design that reduces weight while maintaining strength, and determining effective manufacturing methods for the topologically optimized designs. Efforts are expected to perform comparison testing between topologically optimized and traditionally manufactured components to assess weight, cost, and performance impacts.

Anticipated Benefits: Anticipated benefits for this topic include increased power density (due to reduced weight) and reduced life cycle cost (due to the reduction of material required and increased manufacturing speed).

Engines Topic Descriptions

6--Topic Name: Lightweight and High Strength Compressor Materials
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Preferred Research Focus: Manned and Unmanned Army Rotorcraft Engine Systems

Basis of Need/Capabilities Gap: Army rotorcraft engine compressors typically utilize high strength titanium or nickel-based alloys. These traditional materials are heavy and have relatively low durability from fine sand particle ingestion. The ingestion of fine sand into the compressor leads to airfoil erosion and performance degradation. Erosion barrier coatings (EBCs) are typically employed to prevent leading edge erosion and aerodynamic performance loss, but these increase system cost. Inlet particle separator and filter technologies can be used to reduce sand particles, but these solutions impact overall cycle performance and do not eliminate ingestion of fine sand into the compressor.

Solution Guidelines: TDD seeks development of new lightweight and high strength compressor materials that also have improved durability characteristics. Challenges that must be overcome include optimizing compressor designs to take advantage of new lightweight materials while also achieving increased durability with high-speed rotor components, as well as improving durability without reducing engine performance. Development efforts are expected to include component rig/engine demonstration at relevant scale and cycle parameters of a notional

transition propulsion system. Efforts are also expected to include assessment of system level performance benefits, to clearly support future transition capability.

Anticipated benefits: Anticipated benefits for this topic include improved power density (due to lighter weight materials or reduced parts counts), and improved sustainability (due to high strength materials providing improved durability).

7--Topic Name: Optimized Compressor Aerodynamics

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Engine Systems

Basis of Need/Capabilities Gap: Current Army rotorcraft propulsion systems are limited in compressor Overall Pressure Ratio (OPR) as a result of manufacturing limitations to support production of current aerodynamic design capability, in relation to relative airfoil sizes to clearance closure capability. Army rotorcraft are also limited in various flight maneuvers due to current compressor operability margins to support large accelerations and decelerations. Active tip clearance control technology can be used to manage operability, but these solutions are often heavy and complex. Advanced aerodynamic airfoil shapes are limited by manufacturability of complex geometries in relatively small-scale applications.

Solution Guidelines: TDD seeks technologies that can improve compressor aerodynamics to increase the pressure ratio while also increasing operability margins. Advancements in aerodynamic design of the compressor, enabled by advanced manufacturing, are needed to provide increased OPR. Challenges that must be overcome include reducing the weight and complexity of tip clearance control methodologies while meeting articulation requirements. Development efforts are expected to include component rig/engine demonstration at relevant scale and cycle parameters of a notional transition propulsion system. Efforts are also expected to include assessment of system level performance benefits, to clearly support future transition capability.

Anticipated Benefits: Anticipated benefits for this topic include improved power density (due to increased pressure ratio enabling higher power output), and improved operability margin (through optimization of stall/surge lines).

8--Topic Name: Advanced Combustion Systems

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Engine Systems

Basis of Need/Capabilities Gap: Army rotorcraft combustors are typically made from heavy cobalt or nickel-based alloys using traditional machining processes. The use of these traditional materials and manufacturing methods limit the ability to incorporate optimized designs, which could reduce combustor weight and improve efficiency through increased cycle temperatures. Additive manufacturing can be used, but many of the materials available for additive manufacturing do not meet the required mechanical properties. Limitations in conventional manufacturing of cooling designs prohibit increasing cycle temperature for enhanced cycle performance and designs that extend combustor life and durability. In addition, cooling features in the combustor may be susceptible to degradation when operating in sand and dust environments.

Solution Guidelines: TDD seeks advanced combustor materials and manufacturing methods. Challenges that must be overcome include developing a manufacturing method that can reliably incorporate advanced cooling hole designs, developing a manufacturing method that meets the mechanical properties of traditional machining, developing approaches to mitigate the effects of sand/dust, and developing advanced combustor designs that take advantage of new processes to improve fuel atomization and injection. Development efforts are expected to include component rig/engine demonstration at relevant scale and cycle parameters of a notional transition propulsion system. Efforts are also expected to include assessment of system level performance benefits, to clearly support future transition capability.

Anticipated Benefits: Anticipated benefits for this topic include improved power density (due to increased power capability and reduced weight), reduced specific fuel consumption (due to improved combustion efficiency), and

improved sustainability (due to improved durability, reduced susceptibility to sand/dust and simplified maintenance).

9--Topic Name: High Temperature Tolerant Turbine Materials and Durability Improvements

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Engine Systems

Basis of Need/Capabilities Gap: Army rotorcraft turbines are typically made from heavy nickel-based alloys, and often require thermal barrier coatings, due to their high temperature resistance requirements. Ceramic or Ceramic matrix composite (CMC) materials can provide increased temperature capability but are currently limited to static applications. Current thermal barrier coating solutions have poor durability when in the presence of Calcium-Magnesium-Aluminosilicate (CMAS), which reduces overall engine sustainability. Gas generator turbines are limited by current cooling designs that have a propensity to plug with sand/CMAS, leading to airfoil failure.

Solution Guidelines: TDD seeks development of new high temperature resistant turbine blade materials and associated coatings, disk materials, and shroud materials to enable higher cycle temperatures, as well as advanced airfoil cooling hole designs and thermal barrier coatings that improve durability to CMAS. Challenges that must be overcome include developing a high temperature resistant material and design solutions that are tolerant to high rotational stresses and CMAS intrusion. Development efforts are expected to include component rig/engine demonstration at relevant scale and cycle parameters of a notional transition propulsion system. Efforts are also expected to include assessment of system level performance benefits, to clearly support future transition capability.

Anticipated Benefits: Anticipated benefits for this topic include improved power density (due to increased power output by enabling higher cycle temperatures) and improved sustainability (due to improved high temperature durability and reduced susceptibility to CMAS).

10--Topic Name: Advanced Manufacturing to Improve Turbine Performance
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Preferred Research Focus: Manned and Unmanned Army Rotorcraft Engine Systems

Basis of Need/Capabilities Gap: Current turboshaft engine turbine components are manufactured through a traditional process of investment casting, forging, and Computer Numerical Control (CNC) machining processes. These manufacturing processes limit the ability to fabricate aerodynamically optimized designs that could improve engine efficiency and enhance cooling capability. Additive manufacturing can enable fabrication of highly complex airfoil shapes, but the ability to additively manufacture turbine blades with intricate cooling passages required for Army turboshaft engines has not been demonstrated. Current turbine cooling circuit design is limited by manufacturing capability to incorporate complex geometries relative to Army turboshaft size/space considerations.

Solution Guidelines: TDD seeks advanced manufacturing technologies related to improving turbine blade aerodynamics and internal cooling circuits. Challenges that must be overcome in this area include optimizing both the flow path and cooling flow characteristics, as well as developing manufacturing methods that can reliably incorporate complex cooling passages. Development efforts are expected to include component rig/engine demonstration at relevant scale and cycle parameters of a notional transition propulsion system. Efforts are also expected to include assessment of system level performance benefits, to clearly support future transition capability.

Anticipated Benefits: Anticipated benefits for this topic include reduced specific fuel consumption (due to improved turbine power extraction by optimizing blade aerodynamics), and improved power to weight (due to increased chargeable flow by reducing secondary cooling requirements).

Sequence #25-ATTACH5-B**Hybrid VTOL to Enable Operational Capabilities**

Description: The U.S. Army is soliciting critical data and insights on hybrid vertical take-off and landing (VTOL) and short take-off and landing (STOL) aircraft concepts. The Army is seeking to leverage industry expertise to inform Army decision-making regarding the feasibility, operational impact, cost of adoption, and requirements applicable to hybrid technologies for future Army missions.

Topic Audience: Small Business, Large Business, Academia, HBCU/MI

Department of the Army Research, Development, Test, & Evaluation (RDT&E) Budget Activity (BA): 2, Applied Research

Program Element(s): PE 0602183A / Air Platform Applied Research

Project: DK1 / Air Vehicle Integrated & Alternative Tech (AVIAte)

Discipline(s): Hybrid aviation systems, Vertical/short take-off and landing, Unmanned aircraft

Budget:

FY25	FY26	FY27	Total
\$400K	\$1,300K	\$1,500	\$3,200K

Approximate planning numbers

Multiple awards are anticipated, based upon availability of funds. Two to four awards are anticipated with a period of performance between 12-24 months.

Alterations from Standard Process: N/A

Program Schedule: Project schedule is anticipated to conclude at the end of FY27.

Opportunities for Fundamental Research No.

Government Technical POC: David Friedmann, 757-878-2159, david.s.friedmann.civ@army.mil

1--Topic Name: Hybrid VTOL to Enable Operational Capability

Preferred Research Focus: The Government seeks data to provide insights into the technical feasibility, operational impact, and cost implications of integrated hybrid architectures, technologies, and system designs for vertical take-off and landing (VTOL) and very short take-off and landing (STOL) aircraft platforms configured for relevant Army missions (it is assumed that applicants will have knowledge/ideas in this area). Of particular interest:

- Performance enhancements (range, endurance, payload), efficiencies, novel modes of operation (e.g. quiet approach) or other beneficial attributes enabled by hybrid technologies.
- Cost and operational impact.
- Technical challenges and limitations; physical constraints, maturity and development needs; failure modes and mitigations; integration issues and challenges; benchmark weight, efficiencies, and performance of integrated hybrid power implementation schemes.
- Degree of specialized militarization considerations required for suitable performance of relevant Army missions.

Basis of Need/Capabilities Gap: The Army seeks to leverage on-going Industry investments to assess the unique characteristics of hybrid systems that potentially provide warfighters the ability to conduct operations more efficiently and effectively, or enable new modes of operation that enhance warfighter outcomes. This effort is intended to identify ways in which a hybrid VTOL/STOL solution could bring added value to Army operations.

Solution Guidelines: Provide data, information, and insights into the technology building blocks and costs of hybrid aircraft for Army relevant use cases. Develop and validate study insights through analysis, modeling, test data, and demonstrations. Further guidelines are provided below.

- Information and data are desired to inform further risk mitigation, technology maturation planning, program planning, requirements development, and to inform stakeholders inside and outside of the Government of significant findings and developments.
- Unoccupied aircraft of Group 4/5-size (up to approximately 10,000 lbs. at take-off) with vertical take-off and landing (VTOL) and very short take-off and landing (STOL) aircraft are of primary interest. Data from surrogates may be acceptable if it is directly applicable to aircraft described herein.
- Notional use cases and sizing points range from simple low-cost supply/logistics missions in mostly benign conditions to various tactical missions within a threat environment as a means of extending reach (in time and distance) of onboard or deployable effects.
- The scope of this initial Government-funded effort is anticipated to be primarily analytical. However, it may include system integration and testing depending on individual entry points based on maturity of applicant development efforts, Industry cost share, or future funding (none currently identified).
- Autonomy is a desirable capability; however, autonomy development and maturation are not the focus of this S&T funding.
- System life-cycle affordability has high importance.
- If initial results are promising, a decision may be made to further pursue capability-enhancing ideas.
- For purposes of this Topic, a hybrid aircraft is defined as one that has a propulsive system or architecture that combines multiple elements of energy storage; power production; and or thrust production that may be used simultaneously or separately, either independently or cooperatively. Therefore, supplemental technologies that take advantage of onboard electric power generation, such as electric actuation or advanced mission systems, may be included. Advanced technologies, including fuel cells, may be considered. This definition is provided as a means of conveying intent and openness to different technical solutions. It is not meant to be overly constraining or to hinder innovation. Offerors may propose extensions to this definition for consideration by the Government. Aside from these guidelines, there will not be a government-prescribed, best hybrid system configuration or technology set for this project.
- It is already understood that hybrid technology increases range, payload and mission flexibility compared to all-electric aircraft. Conventional technology (mature, well-known, existing) rotorcraft provide a more relevant comparison baseline.

Goals: The Army desires to define the potential and limitations of hybrid-electric technology as applied to VTOL/STOL aviation to provide positive warfighter outcomes in operational scenarios.

- Leverage Industry development of dual-use technology to enhance the Army's technical knowledge and understanding of hybrid technology for aviation systems in operationally relevant applications.
- Characterize the technical and programmatic risks associated with the transition of various hybrid technologies.
- Demonstrate advancements in Warfighter capabilities and improvements to Army operations.
- Inform the Army's aviation S&T strategy, requirements development, and transition planning.

(End of Summary of Changes)